

MANIPAL SCHOOL OF INFORMATION SCIENCES

Manipal Academy of Higher Education

Linux User and Group Management Lab Assignment

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Part A – User and System Investigation

1. Determine the username of the current account

The following command was used to find the username of the account currently being used.

```
whoami
```

The command displays the username of the current user.

2. Display UID, primary GID and supplementary groups

The following command displays the user ID, group ID and group memberships.

```
id
```

The output contains the UID of the user, the primary GID and the supplementary groups associated with the account.

3. Display all users currently logged in

The following command was used to display the users currently logged in to the system.

```
who
```

Another command that can also be used is:

```
w
```

The output shows the logged-in users and some information about their sessions.

4. Determine what activities the logged-in users are performing

The `w` command was used to check the current activity of logged-in users.

```
w
```

It displays information such as login time, idle time and the current activity or command being used.

5. Examine `/etc/passwd` for the current account

The following command can be used to view the current user's entry from `/etc/passwd`.

```
grep "^$(whoami):" /etc/passwd
```

The entry contains the following fields:

- Username

- UID
- GID
- Home directory
- Login shell

The fields in the file are separated using a colon (:).

6. Count the number of user-account entries

The following command was used to count the entries in `/etc/passwd`.

```
wc -l /etc/passwd
```

The number displayed represents the number of entries present in the file.

7. Display only username and UID

The following command displays only the username and UID fields.

```
cut -d: -f1,3 /etc/passwd
```

Here, `-d:` specifies the colon as the separator and `-f1,3` selects the username and UID fields.

8. Find all accounts having UID 0

The following command was used.

```
awk -F: '$3 == 0 {print $1}' /etc/passwd
```

UID 0 is normally associated with the root account. A user with UID 0 has administrative privileges. Therefore, unauthorized access to a UID 0 account can give an attacker complete control over the system.

9. Compare `/etc/passwd` and `/etc/shadow`

The following commands can be used to inspect the two files.

```
cat /etc/passwd
```

```
sudo cat /etc/shadow
```

The `/etc/passwd` file contains general account information such as username, UID, GID, home directory and login shell. The `/etc/shadow` file contains password-related information, including password hashes.

Password hashes are stored in `/etc/shadow` because this file has restricted permissions. Keeping password information separate from the general account information helps prevent normal users from accessing sensitive password data.

Part B – Creating and Verifying Users

1. Create the required users

The following commands were used to create the three users with home directories.

```
sudo useradd -m alice
sudo useradd -m bob
sudo useradd -m charlie
```

The `-m` option creates a home directory for each user.

2. Verify that the users were added

The following command can be used to check the entries.

```
grep -E '^(alice|bob|charlie):' /etc/passwd
```

This displays the entries for Alice, Bob and Charlie.

3. Determine the UID of each user

The following commands were used to check the UID.

```
id alice
id bob
id charlie
```

The output shows the UID and group information for each user.

4. Determine Alice's home directory and default shell

The following command displays Alice's account information.

```
grep '^alice:' /etc/passwd
```

The home directory and login shell can be identified from the corresponding fields.

5. Verify the home directories

The following command was used to check the `/home` directory.

```
ls -l /home
```

The directories `alice`, `bob` and `charlie` should be visible if they were created successfully.

6. Set passwords for the users

Passwords were set using the following commands.

```
sudo passwd alice  
sudo passwd bob  
sudo passwd charlie
```

The system asks for a password for each account.

7. Switch to Alice using a login shell

The following command can be used to switch to Alice.

```
su - alice
```

After switching, the following command can be used to verify the current user.

```
whoami
```

The output should show:

```
alice
```

To return to the original account:

```
exit
```

Part C – Group Management and Access Control

1. Create the groups

The following commands were used to create the two groups.

```
sudo groupadd security
sudo groupadd developers
```

2. Add Alice and Charlie to the security group

The following commands were used.

```
sudo usermod -aG security alice
sudo usermod -aG security charlie
```

The `-aG` option adds the user to the supplementary group without removing existing group memberships.

3. Add Bob to the developers group

The following command was used.

```
sudo usermod -aG developers bob
```

4. Verify group membership

The following commands can be used to check the groups of each user.

```
id alice
id bob
id charlie
```

These commands display the primary group and supplementary groups of each user.

5. Display the security and developers groups

The following command was used to display the required group entries.

```
grep -E '^(security|developers):' /etc/group
```

The output shows the group names and their members.

6. Add Bob to the security group

Bob can be added to the security group without removing him from the developers group using:

```
sudo usermod -aG security bob
```

The `-aG` option is important because it adds the new group while keeping the existing group membership.

7. Verify Bob's group membership

The following command was used to verify Bob's groups.

```
id bob
```

Bob should now have both `developers` and `security` among his group memberships.